

Open-Source Chip Design and Verification in SkyWater 130nm

Caelan Dunlea

The University of Alabama

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Research Presentation Outline

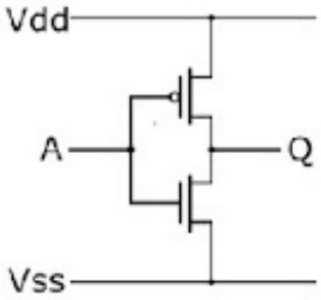
- Motivation
- Tools & Setup
- CMOS Inverter
- 6T SRAM cell
- 2x2 SRAM array
- Cell-ratio tradeoff
- Conclusions

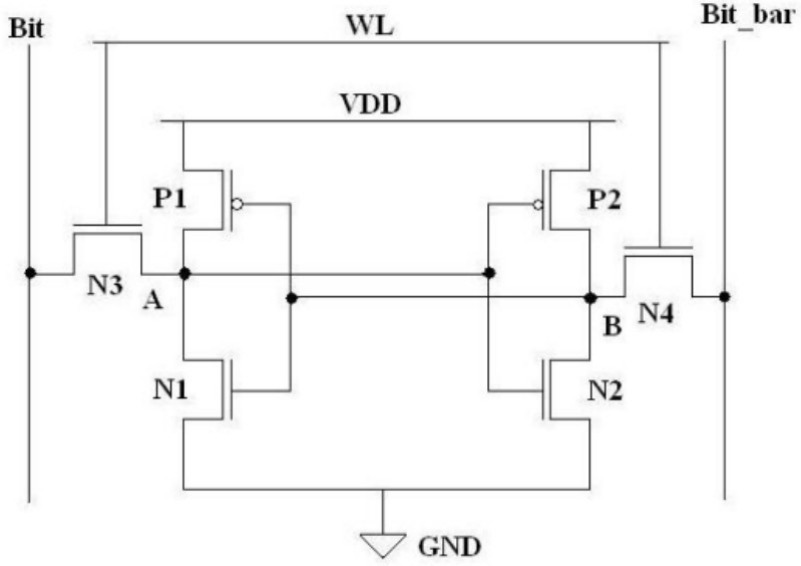
Motivation

- SRAM in high-speed processor caches for CPU
- Caches built from tiling one cell
- Goal: learn full open-source flow on inverter, then build and verify 6T SRAM cell & 2x2 array



Input	Output
1	0
0	1





Open Source Tool Setup

- **Ubuntu VM hosts Xschem, ngspice, Magic, Netgen**
- **Fixed the Xschem old version bug**

```
cadunlea@cadunlea-VirtualBox:~$ tail -n 30 ~/.xschem/simulations/inverter130_schematic.out

Note: No compatibility mode selected!


Circuit: ** sch_path: /home/cadunlea/documents/inverter130/inverter130_schematic.sch

Netlist line no. 5:
Undefined parameter [expr]
Netlist line no. 5:
 Expression err: expr((int((@nf+1)/2)*@w/@nf*0.29))
Netlist line no. 5:
 Formula() error.
      |m=1; mult=1; sd=0; sb=0; sa=0; nf=1; w=1; l=0.15; ad=expr((int((@nf+1)/2)*@w/@nf*0.29)); as=ex
pr((int((@nf+2)/2)*@w/@nf*0.29)); pd=expr((2*int((@nf+1)/2)*(@w/@nf+0.29))); ps=expr((2*int((@nf+2)/2
)*(@w/@nf+0.29))); nrd=expr((0.29/@w )); nrs=expr((0.29/@w ));| : |ad|=|expr((int((@nf+1)/2)*@w/@nf*0
.29))|
Netlist line no. 0:
Undefined parameter [ad]
Netlist line no. 0:
Cannot compute substitute
Netlist line no. 8:
Undefined parameter [expr]
Netlist line no. 8:
 Expression err: expr((int((@nf+1)/2)*@w/@nf*0.29))
Netlist line no. 8:
 Formula() error.
      |m=1; mult=1; sd=0; sb=0; sa=0; nf=1; w=1; l=0.15; ad=expr((int((@nf+1)/2)*@w/@nf*0.29)); as=ex
pr((int((@nf+2)/2)*@w/@nf*0.29)); pd=expr((2*int((@nf+1)/2)*(@w/@nf+0.29))); ps=expr((2*int((@nf+2)/2
)*(@w/@nf+0.29))); nrd=expr((0.29/@w)); nrs=expr((0.29/@w));| : |ad|=|expr((int((@nf+1)/2)*@w/@nf*0.2
9))|
Netlist line no. 0:
```

Before: ngspice expr() errors

```
make[1]: Leaving directory '/home/cadunlea/tools/xschem/src/utile'
cadunlea@cadunlea-VirtualBox:~/tools/xschem$ hash -r && which xschem && xschem -
v
/usr/local/bin/xschem
XSCHEM V3.4.8RC
Copyright (C) 1998-2026 Stefan Schippers

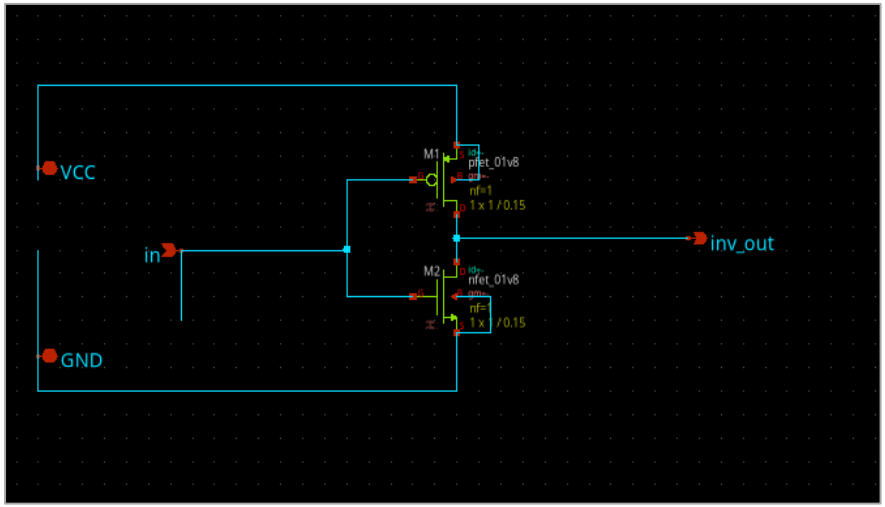
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
Using run time directory XSCHEM_SHAREDIR = /usr/local/share/xschem
Sourcing /usr/local/share/xschem/xschemrc init file

cadunlea@cadunlea-VirtualBox:~/tools/xschem$
```

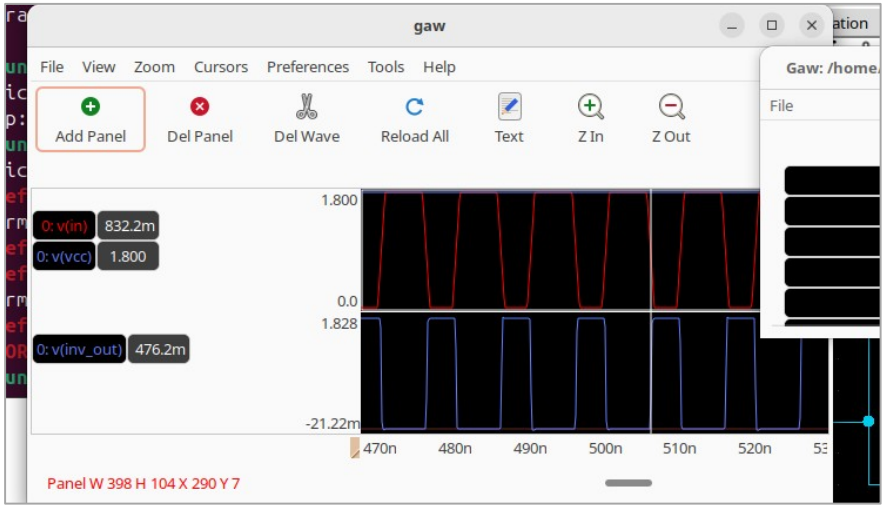
After: Xschem 3.4.8 rebuilt from source

CMOS Inverter Schematic

- PMOS plus NMOS, W=1um, L=0.15um
- Output inverts the input (0 or 1.8V)



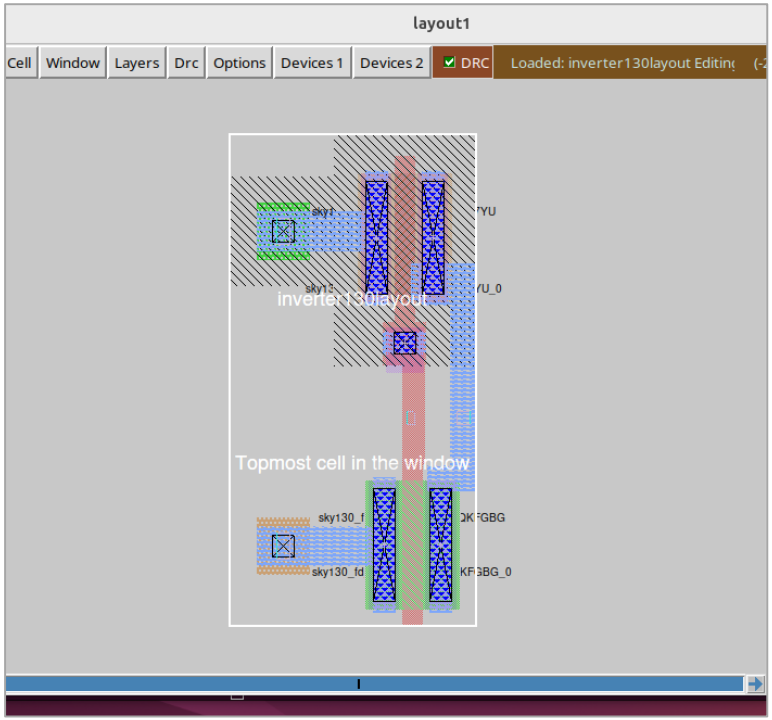
Xschem Schematic



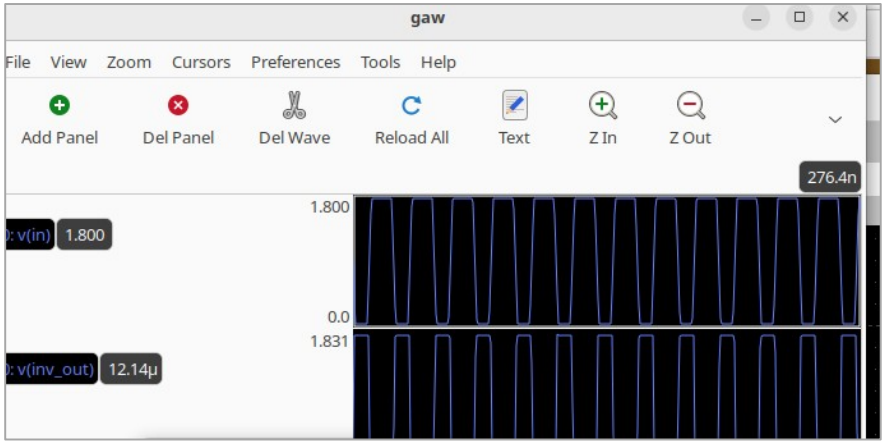
Ngspice waveforms: in (top), out (bottom)

CMOS Inverter Layout & Post-Layout

- 0 DRC errors, LVS “circuits match uniquely”
- Parasitic extraction run, layout still inverts correctly



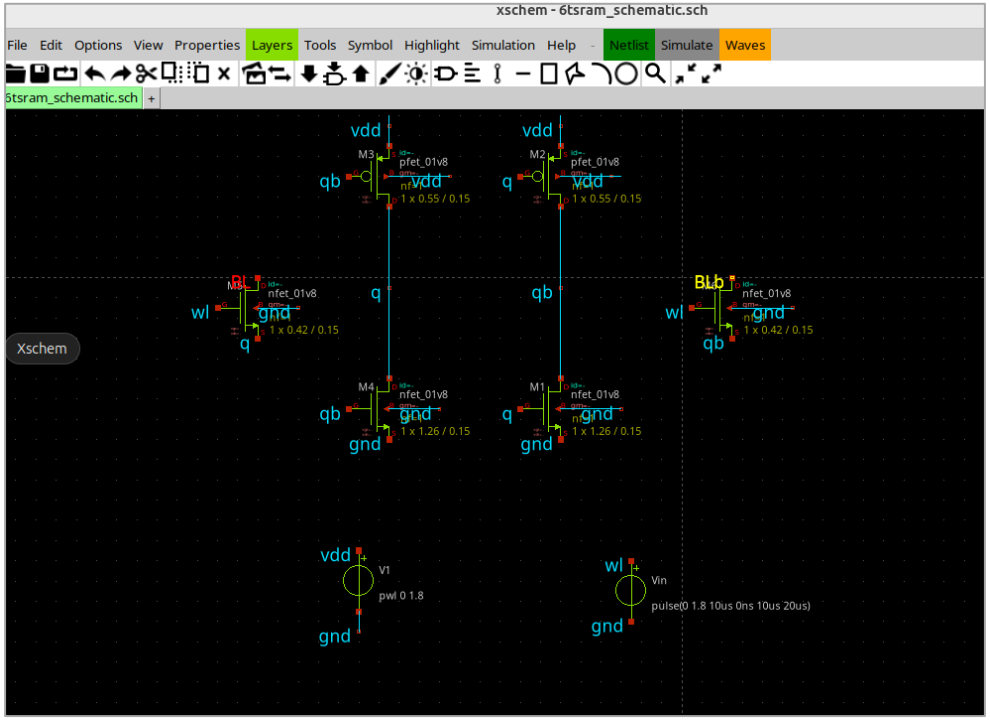
Layout in Magic



Post-layout waveforms

6T SRAM Cell Design

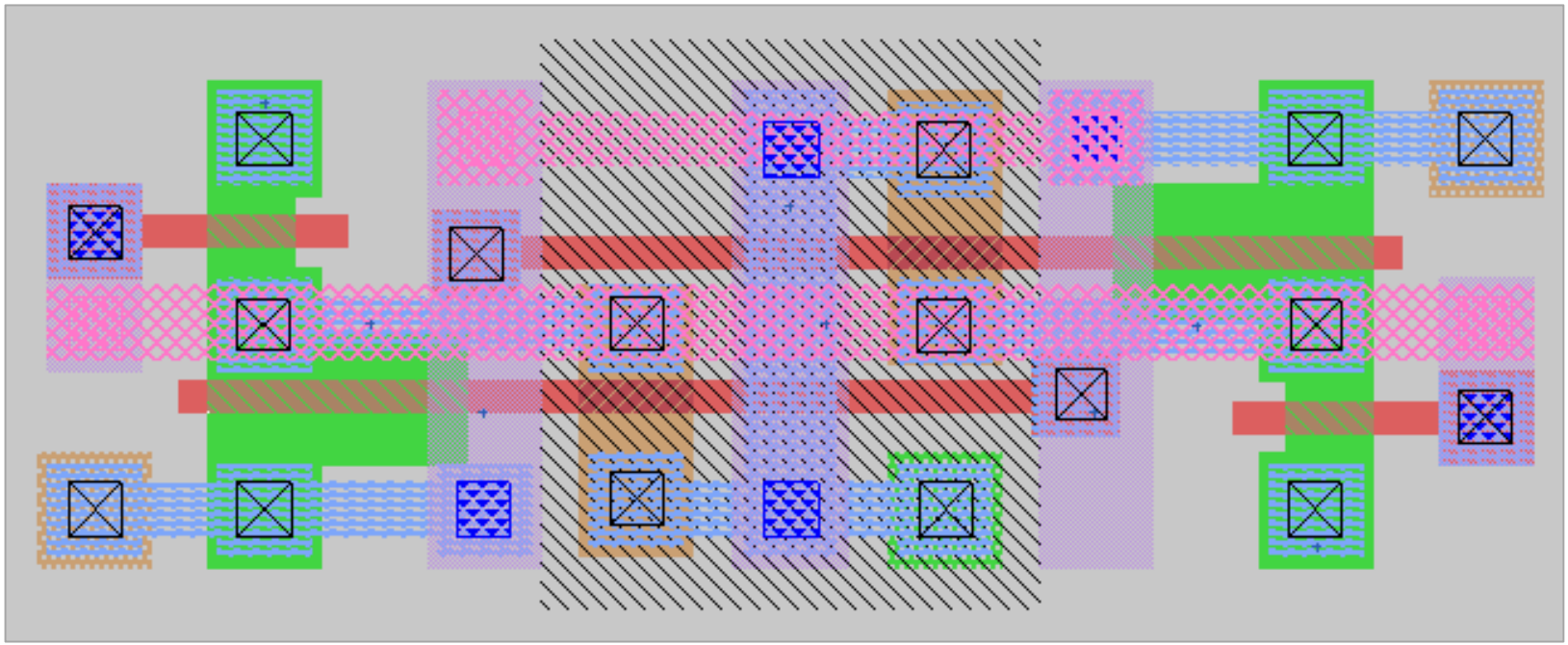
- Two cross-coupled inverters, two access transistors
- Pull-down 1.26um, access 0.42um, pull-up 0.55um



6T cell schematic (cell ratio = 3)

6T Cell Layout & Verification

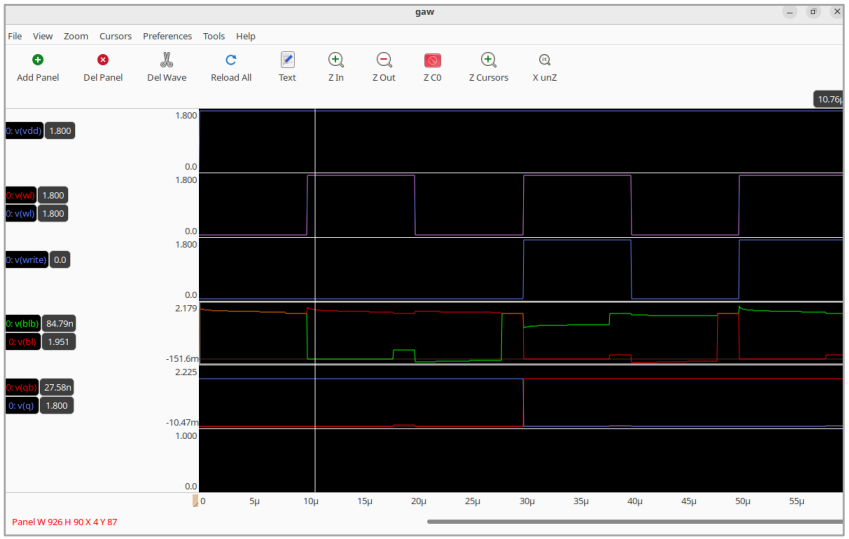
- 0 DRC errors, LVS “circuits match uniquely”
- Pull-down 1.26um, access 0.42um, pull-up 0.55um



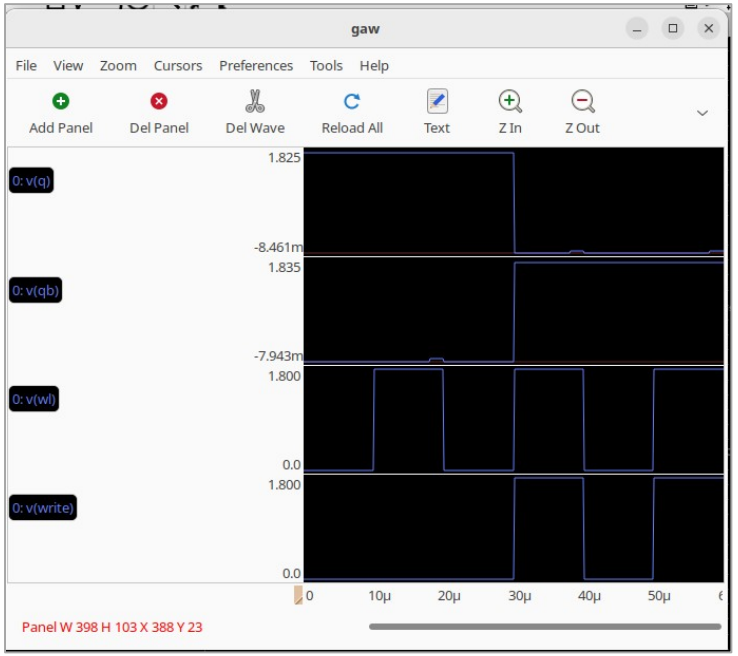
Magic layout: pfets in n-well (hatched), m2 word line (pink)

6T Cell Pre vs Post Layout

- Read (10-20us): q holds, Write (30us): cell fips
- Exhibits same behavior post-pex (18 parasitic capacitors extracted)



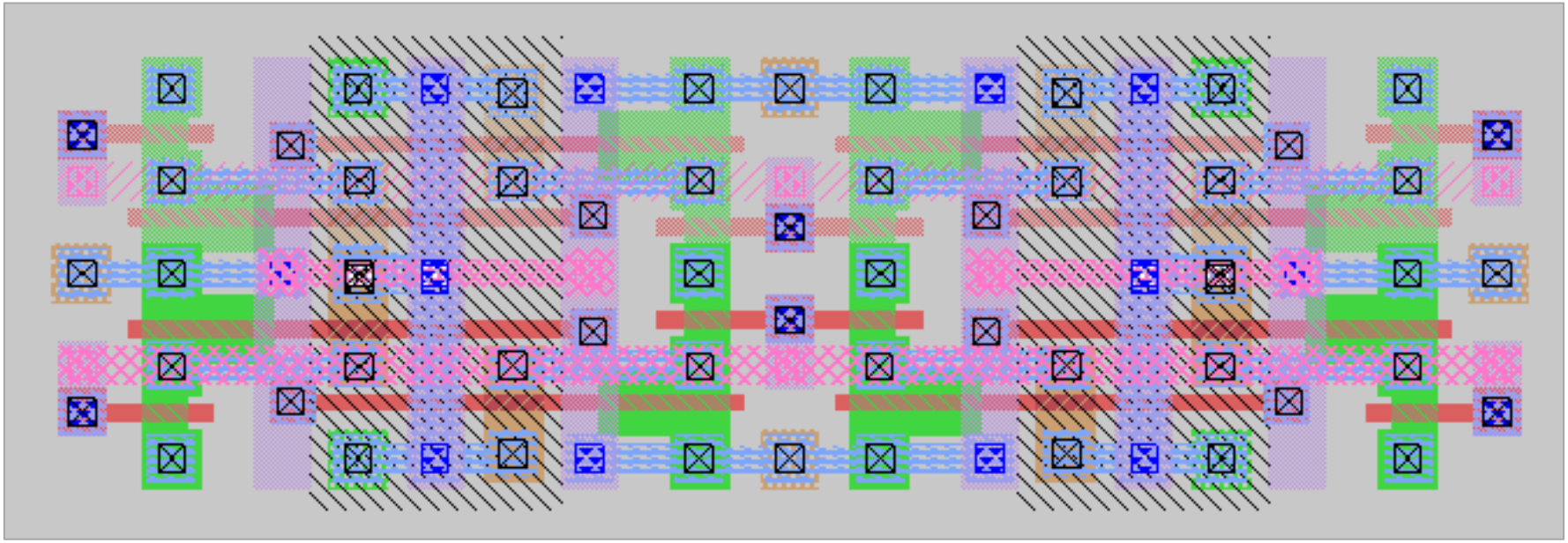
Pre-layout (schematic)



Post-layout (extracted)

2x2 SRAM Array

- Mirrored cells share contacts, reduction in size of 21%
- Array LVS: 24 transistors, “circuits match uniquely”

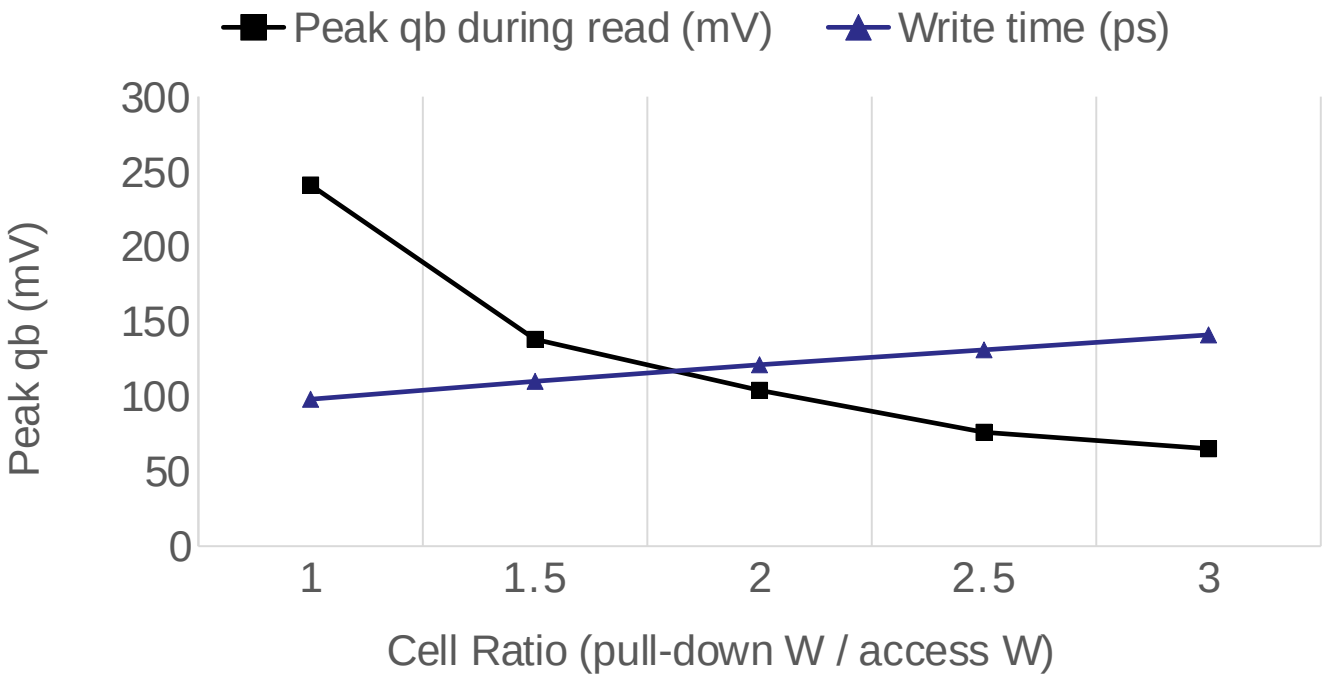


Magic layout: 2x2 array layout

Read vs. Write Trade-off

Performed a cell ratio sweep using ngspice

- **Higher ratio: less read disturb, but slower writes**
- **Ratio of 2-3 balances both**



Conclusions & Future Work

- Completed full open-source design of inverter & SRAM
- Built a DRC LVS clean 6T SRAM cell and 2x2 array
- Compared tradeoffs of read stability against write speed
- Potential future work: static noise margin
- Continue pursuing microelectronics and learning more about VLSI design flows

Thank you!

Questions?